

Pairwise Signature Leakage in UVW-128:

Equivalent Signing-Key Construction and Fresh-Message Forgery from 300 Signatures

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Provenance and review status. The attack was independently found by an AI system, OpenAI Codex using Daybreak Blue at maximum reasoning effort, without a human first proposing this specific signature attack. The AI also produced the derivation, implementation, experiments, and initial manuscript. At the date above, no human cryptanalyst has independently verified the argument, implementation, measurements, or novelty assessment. This preliminary disclosure is provided for human review and reproduction.

Abstract

UVW is a ternary hash-and-sign scheme based on a generalized $(U + V, U + W)$ code. A fixed secret monomial transformation is intended to hide the two public coordinates derived from each secret coordinate. The UVW XY decoder nevertheless leaves a persistent statistical fingerprint on each hidden pair: for a fresh information set of 3,250 secret positions, it forces both coordinates in every selected pair to be nonzero. A signature transmits only the last half of a systematic error, but the first half can be recovered exactly from the public syndrome and public key. Ordinary signatures therefore expose repeated samples of all 9,700 public coordinates.

We score coordinate pairs using a likelihood model derived from the public parameters. Mutual nearest neighbors provide high-confidence pair and relative-scale candidates, and only enough such candidates to span a 1,600-dimensional public subspace are required. Quotienting the public-key columns by this subspace then partitions all 9,700 columns into the 4,850 hidden pairs and determines their relative scales. A public change of syndrome basis exposes matrices of the form

$$\begin{pmatrix} \tilde{H}_X & \tilde{H}_Y \\ -\tilde{H}_Z & \tilde{H}_Z \end{pmatrix}.$$

The submitted Z and XY decoders can then be applied to these matrices, yielding an equivalent signing key without recovering the original seed.

On one UVW-128 key, 300 genuine chosen-message signatures from the submitted reference signer were sufficient. The top 1,600 public candidates spanned the required subspace, the quotient partition uniquely identified every pair, and the resulting decoder produced a 1,244-byte submitted-API signature on a message absent from the query set. The intended mathematical verifier accepted it with weight 8,633. Four parallel signing processes collected the transcript in 47.53 minutes on a four-core Intel Core i5-6500. Public recovery and one forgery took 57.50 seconds and at most 153 MiB. At 200 signatures the same fixed rule failed its public quotient certificate, so no secret-dependent stopping decision is required.

This demonstrates a successful fresh-message forgery for one fixed UVW-128 key. Only one key and one parameter set have undergone end-to-end testing on a genuine source

transcript, so the experiment does not estimate a key-averaged success probability. The attack targets the written sampling method and intended verification equation; it is independent of the separately reported accept-all API wrapper.

1 Introduction

UVW is an NGCC code-based signature submission over $\mathbb{F}_3$ [1]. Its secret parity-check matrix has a two-coordinate block structure, and a fixed monomial map hides both the block matching and the nonzero coordinate scales. Security therefore requires ordinary signatures not to reveal which public coordinates originate from the same secret block.

The signing procedure does not achieve that independence. Its XY decoder selects an information set I of size k_1 . For every $j \in I$, it chooses $e_{1,j}$ so that both entries of

$$(e_{1,j}, e_{1,j} + e_{2,j})$$

are nonzero. The same hidden coordinate pairing is used for every signature under a key. Rejection on the final Hamming weight controls the total number of zeros, but it does not make their locations independent across the two members of a hidden pair.

The attack therefore has a statistical front end followed by an exact algebraic back end. Pairwise likelihood scores first identify a set of reliable pairs. Correctly oriented sums of their public-key columns lie in a fixed subspace U of dimension

$$r_1 = \frac{n}{2} - k_1 = 1600.$$

Once U is known, no further statistical decisions are needed. In the quotient $\mathbb{F}_3^{4850}/U$, the two columns of every hidden pair are the same projective point. The quotient then recovers all remaining pairs and their relative scales, including pairs whose empirical likelihood evidence is weak.

Our contributions are:

- We identify the pairwise distribution leak in the specified UVW signing sampler and show how to reconstruct each complete public error from an ordinary signature.
- We give a public likelihood score for high-confidence pairs and an exact quotient method that completes the entire matching from a basis of U .
- We prove that the recovered pairing and scales yield an equivalent UVW signing key.
- We give a public stopping certificate and distinguish the heuristic likelihood model from the exact algebraic completion.
- We demonstrate the attack end to end on 300 genuine signatures from one submitted UVW-128 key and obtain an intended-verifier-accepted signature on a fresh literal message.

The public UVW signature tracker lists an accept-all wrapper and a verifier crash [3]. Neither issue is used here. We invoke the internal mathematical verifier and require its syndrome and weight checks to pass.

Related UVW-KEM work. A public UVW-KEM reaction-attack lead recovers hidden column pairs and ratios from about 1.06 million precise first-attempt reaction labels, then applies a public projection and generalized Reed–Solomon structure [4, 5]. Its high-volume phase uses secret-assisted simulation, and no black-box reaction classifier is demonstrated. The present signature attack uses the ordinary signing distribution, 300 real source executions, a different 4,850-pair structure, and a quotient completion followed by the signature decoders.

The KEM work is nevertheless structurally related, so we do not claim that this report is the first use of hidden UVW pairs or a cancelling public projection. The signature attack was developed before that KEM report was consulted.

2 UVW-128

Set

$$m = \frac{n}{2} = 4850, \quad r_1 = m - k_1 = 1600, \quad r_2 = m - k_2 = 3250.$$

The UVW-128 parameter tuple is

$$(q, n, k, k_1, k_2, w) = (3, 9700, 4850, 3250, 1600, 8633).$$

Before the secret monomial map, the parity-check matrix has the form

$$H_0 = \begin{pmatrix} H_X & H_Y \\ -H_Z & H_Z \end{pmatrix}, \quad (1)$$

where $H_X, H_Y \in \mathbb{F}_3^{r_1 \times m}$ and $H_Z \in \mathbb{F}_3^{r_2 \times m}$. Key generation applies an invertible row map and a monomial column map, then publishes the systematic matrix

$$H_{\text{pk}} = (I_m \mid R). \quad (2)$$

For a syndrome expressed in the secret row basis as $(s_1 \mid s_2)$, the trapdoor decoder first solves

$$e_2 H_Z^\top = s_2, \quad (3)$$

$$e_1 (H_X + H_Y)^\top = s_1 - e_2 H_Y^\top. \quad (4)$$

The structured error before the monomial map is

$$(e_1 \mid e_1 + e_2). \quad (5)$$

2.1 Every signature exposes the complete public error

A serialized UVW signature contains a salt and only the last $k = m$ entries v of the public error. Let $s = \mathcal{H}(M, r) \in \mathbb{F}_3^m$ be the public hash syndrome. Writing the complete error as $(u \mid v)$, equations (2) and H_{pk} give

$$s = (u \mid v) H_{\text{pk}}^\top = u + v R^\top, \quad u = s - v R^\top. \quad (6)$$

Thus the compact signature does not prevent an attacker from reconstructing all 9,700 public coordinates of the structured error.

3 Pairwise leakage

3.1 A rigorous support fact

Algorithm 5 of the specification chooses an information set $I \subseteq \{1, \dots, m\}$ of size k_1 and requires

$$\text{wt}(e_{1,I}) = \text{wt}((e_1 + e_2)_I) = k_1.$$

The reference implementation realizes this rule by setting

$$e_{1,j} = \begin{cases} e_{2,j}, & e_{2,j} \neq 0, \\ \text{uniform in } \{1, 2\}, & e_{2,j} = 0, \end{cases} \quad j \in I. \quad (7)$$

Both entries of pair j are therefore nonzero whenever $j \in I$. This is an exact support property and does not rely on a random-code approximation. The final fixed-weight rejection can alter other probabilities, but it cannot introduce a zero into a selected pair.

3.2 A heuristic likelihood model

The likelihood model used in the experiment depends only on public parameters. It approximates affine-solved coordinates as uniform and ignores the final weight conditioning. These approximations affect the empirical sample complexity, but not the algebraic certificate applied once a candidate subspace has been found.

Let $\alpha = k_1/m$ and $\beta = 1 - \alpha$. In the submitted Z decoder, a coordinate is among k_2 free positions with probability k_2/m ; the free weight is uniform in $\{0, \dots, k_2\}$, giving mean nonzero probability $1/2$. A solved coordinate is modeled as uniform in $\mathbb{F}_3$. The approximate zero probability of one e_2 coordinate is consequently

$$\zeta = \frac{k_2}{m} \frac{1}{2} + \frac{m - k_2}{m} \frac{1}{3} = 0.3883161512. \quad (8)$$

Normalize the unknown relative scale of a true public pair so that its dominant nonzero relation is equality. The parameter-derived joint model is

$$P \approx \begin{pmatrix} 0.04270143 & 0.03363210 & 0.03363210 \\ 0.03363210 & 0.23857772 & 0.17280736 \\ 0.03363210 & 0.17280736 & 0.23857772 \end{pmatrix}. \quad (9)$$

For the null model we use the public fixed-weight marginal

$$p_X = \left(\frac{n-w}{n}, \frac{w}{2n}, \frac{w}{2n} \right) = (0.11, 0.445, 0.445).$$

For public coordinates a, b and relative scale $\epsilon \in \{1, 2\}$, define

$$L_{a,b}(\epsilon) = \sum_{t=1}^T \log \frac{P(X_a^{(t)}, \epsilon X_b^{(t)})}{p_X(X_a^{(t)}) p_X(\epsilon X_b^{(t)})}. \quad (10)$$

The score of $\{a, b\}$ is the larger of the two values. Each coordinate selects its highest-scoring partner. We retain only mutual selections and sort them by score, avoiding the need to commit immediately to a noisy perfect matching.

4 Exact public completion

4.1 The pair-sum subspace

Let $h_i \in \mathbb{F}_3^m$ denote public column i . After accounting for the hidden monomial map, the two columns belonging to hidden position j can be written

$$a_j = c_j S \begin{pmatrix} x_j \\ -z_j \end{pmatrix}, \quad b_j = d_j S \begin{pmatrix} y_j \\ z_j \end{pmatrix},$$

for nonzero $c_j, d_j \in \mathbb{F}_3$, an invertible row map S , and columns x_j, y_j, z_j of the secret matrices. Hence

$$a_j + \lambda_j b_j = c_j S \begin{pmatrix} x_j + y_j \\ 0 \end{pmatrix} \in U, \quad \lambda_j = c_j d_j^{-1}, \quad (11)$$

where

$$U = S(\mathbb{F}_3^{r_1} \times \{0\}^{r_2}) \subset \mathbb{F}_3^m, \quad \dim U = r_1.$$

The relative scale selected by the likelihood score determines the combination in (11). Correct high-confidence candidates can therefore reveal U by ordinary row reduction.

4.2 Quotient recovery of every pair

Put a row-reduced basis of U into systematic form $[I_{r_1} \mid A]$ after partitioning the ambient coordinates into pivot positions P and free positions F . Define the quotient map

$$Q(v) = v_F - v_P A \in \mathbb{F}_3^{r_2}. \quad (12)$$

Its kernel is exactly U . Equation (11) implies

$$Q(a_j) = -\lambda_j Q(b_j). \quad (13)$$

Normalize each nonzero quotient vector by multiplying its first nonzero entry to one. The two members of a hidden pair then have identical normalized quotients. On the tested public key, the 9,700 columns split into exactly 4,850 normalized classes of size two. The unnormalized quotient vectors also give λ_j .

Once U is correct, this completion stage is exact rather than heuristic: it uses only arithmetic over $\mathbb{F}_3$. Conversely, an incorrect candidate subspace normally produces singleton classes or ambiguous multiplicities, which are visible from public data.

4.3 A target-secret-independent stopping rule

The following procedure uses no secret values:

1. collect signatures in fixed batches and reconstruct their complete public errors with (6);
2. compute mutual likelihood candidates with (10);
3. take the first $r_1 = 1600$ candidates and require their oriented sums to have rank r_1 ;
4. construct Q and require every normalized quotient class to contain exactly two public columns;
5. recover the equivalent matrices, decode a fresh hash syndrome, and accept the result only if the intended public verifier accepts.

If any condition fails, the attacker collects the next batch and repeats the procedure. The experiment used checkpoints at 200 and 300 signatures. The 200-signature checkpoint reached rank 1,600 but failed the quotient partition with a singleton class. The 300-signature checkpoint passed every condition. The stopping decision therefore did not inspect the target secret.

5 Equivalent signing-key construction

Theorem 1 (Equivalent UVW form). *Assume that U , the hidden pairs, and their relative scales have been recovered as above. From public information one can construct an invertible row map T and matrices*

$$\tilde{H}_X, \tilde{H}_Y \in \mathbb{F}_3^{r_1 \times m}, \quad \tilde{H}_Z \in \mathbb{F}_3^{r_2 \times m}$$

such that, after reordering and relatively scaling paired columns,

$$T H_{\text{pk}}^* = \begin{pmatrix} \tilde{H}_X & \tilde{H}_Y \\ -\tilde{H}_Z & \tilde{H}_Z \end{pmatrix}. \quad (14)$$

Moreover, $\tilde{H}_X + \tilde{H}_Y$ has row rank r_1 and $\tilde{H}_Z$ has row rank r_2 .

Proof. Use the pivot/free coordinates above and define

$$T(v) = (v_P, Q(v)).$$

This map is invertible because v is recovered from v_P together with $v_F = Q(v) + v_P A$. Scale the second member of every recovered pair so that its sum with the first lies in U . Their quotient components are then opposites by (13). The bottom r_2 rows consequently have the form $(-\tilde{H}_Z \mid \tilde{H}_Z)$, after a harmless common scale and pair ordering. The top rows define $(\tilde{H}_X \mid \tilde{H}_Y)$.

The pair sums span U , so their top coordinates have rank r_1 ; these are the columns of $\tilde{H}_X + \tilde{H}_Y$. The public matrix has full row rank, hence its column images span $\mathbb{F}_3^m/U$ in the quotient. Those images are the columns of $\tilde{H}_Z$ up to signs and scales, so $\tilde{H}_Z$ has rank r_2 . $\square$

For a target public syndrome s , compute $T(s) = (\tilde{s}_1 \mid \tilde{s}_2)$. Apply the published decoder equations

$$\begin{aligned} e_2 \tilde{H}_Z^\top &= \tilde{s}_2, \\ e_1 (\tilde{H}_X + \tilde{H}_Y)^\top &= \tilde{s}_1 - e_2 \tilde{H}_Y^\top. \end{aligned}$$

When the ordinary weight rejection returns a vector of weight w , map $(e_1 \mid e_1 + e_2)$ back through the recovered public pairs and scales.

Corollary 2 (Fresh-message forgery). *The recovered matrices and column map form an equivalent signing key. They decode the hash syndrome of an arbitrary message and salt, and the resulting signature satisfies the intended UVW verification equation.*

The construction recovers an equivalent decoder rather than the original key seed, nonces, or necessarily the submitter’s exact secret matrices.

6 End-to-end experiment

6.1 Setup and separation from secret diagnostics

The experiment used the archived UVW-128 reference implementation whose public tracker archive hash is

bbfa8dad5ee57083578b50b9937e773e6158f72646e825da3d3265cc00cb1294.

The signing and decoding logic was left unchanged. Pre-existing diagnostic prints were removed, and a key-generation helper retained the reconstructed key in memory so that a separate ground-truth file could be written for later evaluation. The public attack itself read only the public key and the reconstructed-error transcript.

The ground-truth map was consulted only after recovery to measure pair accuracy. Running the estimator with diagnostic evaluation enabled and disabled produced byte-identical candidate files with SHA-256

c64058d33637865c223ba7d8556b6bbcfce80b2cc92bbf9236642decafb8ba9a.

Thus access to ground truth did not affect partner selection, relative-scale selection, the stopping rule, or the forgery.

All timings were measured on Linux with a four-core Intel Core i5-6500 at 3.20 GHz and 16 GiB of RAM. Key generation took 132.794 seconds and is not part of the attack cost because the challenger supplies the public key.

6.2 Transcript collection

Four signing processes generated 75 distinct chosen-message signatures each. The slowest process completed in 2,851.507 seconds, giving a transcript wall time of 47.53 minutes. Each signature was generated by the submitted signer. Its salt and message hash were used to reconstruct the systematic head with (6); every reconstructed error had the required weight 8,633.

Each child process reinitialized its process-local submission DRNG after the fork. Worker $w \in \{0, 1, 2, 3\}$ used the distinct 64-byte seed

$$\text{seed}_w[i] = (167 + 41i + 73w) \bmod 256, \quad 0 \leq i < 64.$$

These are fixed reproducibility seeds, not four draws from an external entropy source. The submitted SM3-based derivation consumes all 64 bytes and gave four distinct DRNG states. The salt, the Z -decoder information set and free values, and the XY -decoder information set and free values all draw from the corresponding child-local state. Thus the four workers used separate pseudorandom streams rather than cloned post-fork state. As checks against accidental stream reuse, all 300 reconstructed errors were distinct, and the pooled same-index cross-worker Pearson correlations of their binary nonzero indicators lay between $-8.11 \cdot 10^{-4}$ and $2.04 \cdot 10^{-3}$.

The public key and 300-error transcript have SHA-256 digests

```
ea4978ef4c7afd3dcfcf7bc70ca39830e7f853ccabe9d9bde037f12bc23390ba,
5c44e89bf891ebca8c3588f9c498f561593259b963a1ceda24b7f8002d1f6287,
```

respectively.

6.3 Recovery and certificate

Table 1 reports the public checkpoint results. The “correct seeds” column uses ground truth only for retrospective evaluation.

Table 1: Fixed public recovery rule at transcript checkpoints.

T	mutual pairs	correct seeds	seed rank	public outcome
200	3,118	1,544	1,600	reject: singleton quotient
300	4,074	1,600	1,600	accept: 4,850 quotient pairs

At 300 signatures, the complete mutual list contained 3,992 true pairs among 4,074 candidates. The top 2,000 were all correct, although the attack used only the first 1,600. Their oriented public-column sums spanned a subspace of dimension exactly 1,600. The quotient recovered all 4,850 pairs and their relative scales. A post-recovery comparison with the withheld map confirmed every pair and scale.

The likelihood stage took 16.74 seconds, 15.80 user CPU-seconds, and 156,516 KiB peak resident memory. With the fixed 1,600-candidate rule, subspace construction, quotient recovery, equivalent matrices, and one decode took 40.76 seconds, 14.02 user CPU-seconds, and 54,632 KiB peak resident memory. Total public offline wall time was 57.50 seconds, with peak memory below 153 MiB.

6.4 Fresh-message forgery

The recovered equivalent key was used to sign the literal message

```
fresh UVW-128 message forged from public transcripts.
```

This message was absent from the 300-query set. The submitted API encoding is 1,244 bytes: 12 parameter bytes, a 16-byte salt, and 1,216 bytes for the two bit-slices of the 4,850-trit tail. The signature passed the internal `uvw_verify` predicate, and its reconstructed full error had weight 8,633. The API byte string has SHA-256

`de42607a59044ca91b647d610be24a601b2cb6413c5f20971cd39f3688857241.`

The 1,252-byte experimental evidence file prepends the internal eight-byte `UVWSSIGN` file header; that container has SHA-256

`cb6c9fa7a81cf56314369bd718a076da1f5d9b89cc0c2086ef0d803c4182787f.`

The same API signature and evidence file were obtained with the earlier 2,000-candidate diagnostic run and the fixed 1,600-candidate execution.

7 What is rigorous and what is empirical

The following parts of the attack are exact:

- reconstruction of the full public error from every signature;
- the forced-nonzero condition on each selected hidden pair;
- membership of correctly scaled pair sums in U ;
- quotient completion once U is known;
- Theorem 1 and the resulting decoder; and
- acceptance of the fresh-message forgery by the intended verifier.

The likelihood distribution in (9) is heuristic. It models solved coordinates as uniform and does not derive the exact joint law after fixed-weight rejection and information-set invertibility conditioning. Its role is only to supply enough correct candidates to span U . The observed 300-signature threshold is likewise empirical and may vary across keys. The public quotient partition and final verification provide public checks against silently accepting an incorrect statistical guess.

8 Security implication and scope

The signing queries may use arbitrary distinct messages. After the equivalent signing key is constructed, the adversary chooses a fresh message and salt, hashes them, decodes the new syndrome, and returns the serialized tail and salt. This has the form of an EUF-CMA attack. The concrete experiment demonstrates a successful fresh-message forgery for one fixed UVW-128 key.

Only one random key was subjected to recovery from a genuine source transcript, and only UVW-128 was tested end to end. The attack method was developed while analyzing that instance. A target-secret-independent rerun on the same transcript used the fixed r_1 candidate count and public quotient stopping certificate, but it is not a prospective new-key trial. No success-rate claim over random key generation is made.

UVW-256 and UVW-512 use the same forced-pair mechanism. The algebraic completion extends directly, while their signature requirements and practical running times have not been measured. They are therefore not claimed broken in this report.

The attack concerns both the specified sampler and the submitted reference instantiation. In particular, the source instantiates the Z -decoder weight choice uniformly over $\{0, \dots, k_2\}$, which is used in the heuristic score. A different distribution would require a new likelihood model, but it would not remove the exact forced-nonzero fact in Algorithm 5.

The security proof simulates signatures with vectors sampled uniformly from the public Hamming sphere. The specification’s domain-sampling results analyze syndromes of vectors already drawn from that distribution. They do not establish that the trapdoor decoder’s output is close to that same distribution. The persistent pair test distinguishes the submitted decoder output from a uniform fixed-weight vector, so this simulation step requires a new argument or a redesigned sampler.

9 Countermeasures

Changing the fixed monomial permutation or its signs changes only the unknown matching that the attack must learn. Increasing the dimensions enlarges the polynomial-time scoring and quotient computations without removing the source of leakage.

A repair requires a preimage sampler whose output distribution is proven to be close to a public, key-independent target distribution across many signatures. Rejection on total Hamming weight is insufficient when the secret decoder forces both members of thousands of fixed hidden pairs to be nonzero. A revised proof should compare the actual decoder output with the exact signing-oracle distribution used in the EUF-CMA games, and full-size tests should include all pairwise public-coordinate contingency tables under one fixed key.

10 Conclusion

UVW-128’s monomial map hides coordinate labels, but it does not hide the repeated statistical dependence created by the XY decoder. For the tested key, three hundred ordinary signatures exposed enough high-confidence pairs to span a 1,600-dimensional public subspace. Quotienting by that subspace recovers the entire hidden matching and relative scales exactly. The resulting matrices form an equivalent UVW signing key and produce a fresh-message signature accepted by the intended verifier.

In the demonstrated run, query collection took under an hour and the public offline computation took under one minute on an older four-core desktop. Its statistical sample threshold remains a one-key observation, while the algebraic completion and final forgery are publicly checkable.

References

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